Physics-Informed Artificial Neural Network Framework
for Battery State of Health (SOH) and
Remaining Useful Life (RUL) Prediction
Using Thermal Gradient and Stress-Based Degradation Modelling

Methodology Chapter

NASA Ames Prognostics Centre of Excellence Dataset

This chapter presents the complete methodology for a hybrid Physics-Informed Neural Network (PINN) system for lithium-ion battery health prognostics.

Nomenclature

Symbol	Description	Units
SOH	State of Health	–
RUL	Remaining Useful Life	cycles
PDI	Physics Degradation Index	–
T_{core}	Core temperature of the battery cell	K
T_{surf}	Surface temperature of the battery cell	K
ΔT	Thermal gradient ($T_{\text{core}} - T_{\text{surf}}$)	K
σ	Thermal stress	MPa
Q	Measured discharge capacity	A h
Q_0	Nominal (rated) capacity	A h
I	Applied current	A
V	Terminal voltage	V
R_{int}	Internal resistance	Ω
$\dot{q}_{\text{gen}}$	Volumetric heat generation rate	$\mathrm{W\,m^{-3}}$
k	Thermal conductivity	$\mathrm{W\,m^{-1}\,K^{-1}}$
ρ	Mass density	$\mathrm{kg\,m^{-3}}$
c_p	Specific heat capacity	$\mathrm{J\,kg^{-1}\,K^{-1}}$
h	Convective heat transfer coefficient	$\mathrm{W\,m^{-2}\,K^{-1}}$
E_a	Activation energy (Arrhenius degradation)	$\mathrm{J\,mol^{-1}}$
R_g	Universal gas constant	$\mathrm{J\,mol^{-1}\,K^{-1}}$
k_d	Degradation rate constant	–
$\lambda_1, \ldots, \lambda_4$	Loss weighting hyperparameters	–
$\mathbf{W}^{(l)}$	Weight matrix of layer l	–
$\mathbf{b}^{(l)}$	Bias vector of layer l	–
$\mathbf{z}^{(l)}$	Pre-activation vector at layer l	–
$\mathbf{h}^{(l)}$	Post-activation (hidden representation) at layer l	–
$f(\cdot)$	Activation function	–
N	Cycle number	cycles
N_{EOL}	End-of-life cycle number	cycles
α_T	Coefficient of thermal expansion	$\mathrm{K^{-1}}$
E_Y	Young's modulus	GPa
Σ_{cum}	Cumulative thermal stress index	–
$\mathcal{L}$	Total physics-informed loss	–

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Chapter 1

Problem Formulation

1.1 Battery State of Health

The *State of Health* (SOH) quantifies the current energy-storage capability of a lithium-ion cell relative to its rated nominal capacity. It is formally defined as:

$$\mathrm{SOH}(N) = \frac{Q(N)}{Q_0} \times 100\% \tag{1.1}$$

where $Q(N)$ is the measured discharge capacity at cycle N and Q_0 is the nominal factory capacity. An SOH of 100% represents a new cell; a cell is typically declared at its *End of Life* (EOL) when SOH $\leq 80\%$ [1].

1.2 Remaining Useful Life

The *Remaining Useful Life* (RUL) denotes the number of charge–discharge cycles remaining before the cell reaches its EOL threshold:

$$\mathrm{RUL}(N) = N_{\mathrm{EOL}} - N \tag{1.2}$$

where N_{EOL} is the cycle at which $Q(N_{\mathrm{EOL}})/Q_0 \leq 0.80$.

1.3 Capacity Fade and Degradation

Capacity fade is modelled as a monotonically decreasing function of cycle number. A common empirical form is:

$$Q(N) = Q_0 \exp\left(-k_d \cdot N^{\beta}\right) \tag{1.3}$$

where k_d is a degradation rate constant and β is an empirically fitted exponent ($\beta \approx 0.5$–1 for lithium-ion chemistry). The instantaneous degradation rate is:

$$\frac{\mathrm{d}Q}{\mathrm{d}N} = -k_d \beta N^{\beta-1} Q_0 \exp\left(-k_d N^{\beta}\right) \tag{1.4}$$

An Arrhenius-type temperature dependence further modifies k_d:

$$k_d(T) = A_0 \exp\left(-\frac{E_a}{R_g T}\right) \tag{1.5}$$

where A_0 is a pre-exponential factor, E_a is the activation energy, and $R_g = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}$.

Chapter 2

Analytical Physics Model

2.1 Heat Generation

During charge–discharge cycling, irreversible (Joule) and reversible (entropic) heat is generated within the cell. The total volumetric heat generation is:

$$\dot{q}_{\text{gen}} = \dot{q}_{\text{Joule}} + \dot{q}_{\text{entropy}} \tag{2.1}$$

2.1.1 Joule Heating

$$\dot{q}_{\text{Joule}} = I^2 R_{\text{int}} \tag{2.2}$$

2.1.2 Entropic Heat Generation

$$\dot{q}_{\text{entropy}} = I \cdot T \cdot \frac{\partial U_{\text{OCV}}}{\partial T} \tag{2.3}$$

where U_{OCV} is the open-circuit voltage. For simplicity in many applications, $\dot{q}_{\text{entropy}} \ll \dot{q}_{\text{Joule}}$ is assumed, reducing to $\dot{q}_{\text{gen}} \approx I^2 R_{\text{int}}$.

2.2 Thermal Energy Balance

The transient heat equation governing the cell temperature field $T(\mathbf{r}, t)$ is:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + \dot{q}_{\text{gen}} \tag{2.4}$$

Boundary condition (convective cooling):

$$-k \nabla T \cdot \hat{n}\big|_{\text{surf}} = h(T_{\text{surf}} - T_\infty) \tag{2.5}$$

2.3 Lumped Thermal Model

Assuming one-dimensional radial symmetry in a cylindrical cell, a two-node lumped model separates core and surface temperatures:

$$m_c c_p \frac{\mathrm{d}T_{\text{core}}}{\mathrm{d}t} = \dot{q}_{\text{gen}} - \frac{T_{\text{core}} - T_{\text{surf}}}{R_{\text{cond}}} \tag{2.6}$$

$$m_s c_p \frac{\mathrm{d}T_{\text{surf}}}{\mathrm{d}t} = \frac{T_{\text{core}} - T_{\text{surf}}}{R_{\text{cond}}} - \frac{T_{\text{surf}} - T_{\infty}}{R_{\text{conv}}} \tag{2.7}$$

where $R_{\text{cond}} = 1/(4\pi k l)$ and $R_{\text{conv}} = 1/(hA_s)$ are conductive and convective thermal resistances; m_c, m_s are effective masses; l is the cell length.

2.4 Thermal Gradient

The key physics feature is the temperature differential:

$$\Delta T(t) = T_{\text{core}}(t) - T_{\text{surf}}(t) \tag{2.8}$$

The temporal rate of change:

$$\frac{\mathrm{d}(\Delta T)}{\mathrm{d}t} = \frac{\mathrm{d}T_{\text{core}}}{\mathrm{d}t} - \frac{\mathrm{d}T_{\text{surf}}}{\mathrm{d}t} \tag{2.9}$$

An elevated ΔT at a given current is indicative of increased internal resistance and thus accelerated degradation.

2.5 Physics-Based SOH and RUL Estimators

The Physics Degradation Index (PDI) integrates thermal stress over the operational history:

$$\text{PDI}(N) = \frac{1}{N} \sum_{n=1}^{N} \frac{\sigma_{\text{cum}}(n)}{\sigma_{\text{ref}}} \tag{2.10}$$

The physics-based SOH estimator combines capacity fade with the PDI:

$$\text{SOH}_{\text{phys}}(N) = (1 - k_d \cdot N)\,(1 - \gamma\,\text{PDI}(N)) \tag{2.11}$$

The physics-based RUL is obtained by solving $\text{SOH}_{\text{phys}}(N + \text{RUL}) = 0.80$:

$$\text{RUL}_{\text{phys}}(N) = \frac{1 - 0.80/(1 - \gamma\,\text{PDI}(N))}{k_d} - N \tag{2.12}$$

These serve as the physics reference signals that regularise the neural network.

Chapter 3

Temperature Differential Model

3.1 Motivation

The temperature difference ΔT between cell core and surface is a sensitive early indicator of degradation. As the internal resistance R_{int} increases with ageing, Joule heating $I^2 R_{\text{int}}$ rises while the cell's thermal conductance decreases, resulting in a larger ΔT for identical operating conditions. Monitoring ΔT therefore allows degradation to be detected before visible capacity fade.

3.2 Differential Temperature Equations

Let $\tau_c = m_c c_p R_{\text{cond}}$ and $\tau_s = m_s c_p R_{\text{conv}}$ be the thermal time constants. From eqs. (2.6) and (2.7):

$$\Delta T(t) = \dot{q}_{\text{gen}} R_{\text{cond}} \left(1 - e^{-t/\tau_c}\right) e^{-t/\tau_s} \tag{3.1}$$

At thermal steady state ($t \to \infty$ within a cycle):

$$\Delta T_{\text{ss}} = \dot{q}_{\text{gen}} \cdot R_{\text{cond}} = I^2 R_{\text{int}} \cdot R_{\text{cond}} \tag{3.2}$$

3.3 Thermal Accumulation Index

Integrating ΔT over a full discharge period $[0, t_d]$:

$$\Xi(N) = \int_0^{t_d} \Delta T(t, N)\, \mathrm{d}t \tag{3.3}$$

The per-cycle thermal accumulation index $\Xi(N)$ monotonically increases with ageing, making it a robust prognostic feature.

3.4 Temperature Stress Feature

A composite temperature stress feature combines gradient and rate:

$$f_T(N) = w_1 \Delta T(N) + w_2 \left.\frac{\mathrm{d}(\Delta T)}{\mathrm{d}t}\right|_{\text{peak}} + w_3 \Xi(N) \tag{3.4}$$

where w_1, w_2, w_3 are normalisation weights tuned on validation data.

Chapter 4

Thermal Stress Model

4.1 Thermo-Mechanical Stress Formulation

Temperature gradients within the electrode stack generate thermo-mechanical stresses that accelerate solid-electrolyte interphase (SEI) growth and particle cracking. The principal thermal stress is:

$$\sigma = E_Y \alpha_T \Delta T \tag{4.1}$$

For a constrained cylindrical geometry, the von Mises equivalent stress is:

$$\sigma_{\text{vm}} = \frac{E_Y \alpha_T \Delta T}{\sqrt{3}(1-\nu)} \tag{4.2}$$

where ν is Poisson's ratio.

4.2 Cumulative Thermal Fatigue

Adopting a Miner's-Rule analogy for thermal cycling damage:

$$D_{\text{Miner}}(N) = \sum_{n=1}^{N} \frac{d_n}{D_f} \tag{4.3}$$

where d_n is the damage increment per cycle and D_f is the failure damage threshold. The incremental stress damage per cycle is:

$$d_n = \left(\frac{\sigma_n}{\sigma_{\text{ult}}}\right)^m \tag{4.4}$$

with material constants σ_{ult} (ultimate stress) and m (fatigue exponent, typically $m = 2$–4).

4.3 Stress Accumulation Model

The cumulative stress index over N cycles is:

$$\Sigma_{\text{cum}}(N) = \sum_{n=1}^{N} \sigma_n = \sum_{n=1}^{N} E_Y \alpha_T \Delta T(n) \tag{4.5}$$

4.4 Physics Degradation Score (PDI)

Combining thermal fatigue damage with normalised stress accumulation:

$$\mathrm{PDI}(N) = \alpha_D \cdot D_{\mathrm{Miner}}(N) + (1 - \alpha_D) \cdot \frac{\Sigma_{\mathrm{cum}}(N)}{\Sigma_{\mathrm{ref}}} \tag{4.6}$$

where $\alpha_D \in [0, 1]$ controls the relative contribution of fatigue damage versus cumulative stress. A PDI of 1.0 denotes EOL; values above 0.9 trigger maintenance alerts.

Chapter 5

Dataset and Feature Engineering

5.1 NASA Prognostics Dataset

The primary dataset is the *NASA Ames Prognostics Centre of Excellence Lithium-Ion Battery Dataset* [1], comprising 18650-format lithium-ion cells ($LiCoO_2$ chemistry) cycled under controlled laboratory conditions. Key dataset characteristics are summarised in table 5.1.

Table 5.1: NASA Ames Battery Dataset summary

Parameter	**Value / Range**	**Unit**
Cell format	18650 cylindrical	–
Nominal capacity Q_0	2.0	Ah
Nominal voltage	3.6	V
Charge protocol	CC-CV at 1.5 A, cutoff 4.2 V	–
Discharge protocol	CC at 2.0 A, cutoff 2.5 V	–
Cycle range	0 – 300+	cycles
Measurement frequency	Per-cycle aggregated	–
Variables recorded	V, I, T, Q, R_{int}	–
Cells available	B0005, B0006, B0007, B0018	–

The methodology generalises to the **CALCE Battery Dataset** (University of Maryland), the **Oxford Battery Degradation Dataset**, and the **Stanford Battery Dataset** by adapting chemistry-specific parameters.

5.2 Raw Feature Set

The raw input features extracted directly from cycling data are listed in table 5.2.

Table 5.2: Raw features extracted from the NASA dataset

Feature	**Symbol**	**Description**
Voltage	V	End-of-discharge terminal voltage
Current	I	Applied discharge current
Capacity	Q	Measured discharge capacity per cycle
Cycle number	N	Cumulative charge–discharge cycle index
Ambient temperature	T_∞	Environmental temperature during cycling
Charge time	t_{chg}	Duration of the constant-current charge phase
Discharge time	t_{dis}	Duration of the discharge phase
Internal resistance	R_{int}	Impedance-spectroscopy or pulse-measured resistance

5.3 Derived Physics Features

The physics engine processes raw features to produce physics-informed derived features (table 5.3).

Table 5.3: Derived physics-based features

Feature	**Symbol**	**Physical significance**
Core temperature	T_{core}	Estimated via lumped thermal model (eq. (2.6))
Surface temperature	T_{surf}	Estimated via lumped thermal model (eq. (2.7))
Thermal gradient	ΔT	Key degradation indicator (eq. (2.8))
Temperature rate	$\mathrm{d}T/\mathrm{d}t$	Rate of thermal change within cycle
Thermal stress	σ	Thermo-mechanical stress (eq. (4.1))
PDI	PDI	Physics Degradation Index (eq. (4.6))
Physics SOH	$\mathrm{SOH}_{\mathrm{phys}}$	Model-based SOH estimate (eq. (2.11))
Physics RUL	$\mathrm{RUL}_{\mathrm{phys}}$	Model-based RUL estimate (eq. (2.12))

Chapter 6

Data Preprocessing

6.1 Missing Value Handling

Missing values arising from sensor dropout or logging errors are handled using cycle-to-cycle linear interpolation:

$$x_{\text{interp}}(N) = x(N_a) + \frac{N - N_a}{N_b - N_a}\big(x(N_b) - x(N_a)\big) \tag{6.1}$$

Features with $> 5\%$ missing data are flagged, and the corresponding cycles are excluded from training.

6.2 Outlier Removal via Z-Score Filtering

For each feature x, the Z-score is computed:

$$z_i = \frac{x_i - \bar{x}}{s_x} \tag{6.2}$$

Samples with $|z_i| > 3$ are treated as outliers and replaced by the cycle-window median.

6.3 Normalisation

6.3.1 Min-Max Scaling

$$\tilde{x}_i = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \tag{6.3}$$

Applied to bounded features such as SOH $\in [0, 1]$ and $N/N_{\max}$.

6.3.2 Standard (Z-score) Scaling

$$\hat{x}_i = \frac{x_i - \bar{x}}{s_x} \tag{6.4}$$

Applied to unbounded features such as voltage, current, and resistance to ensure unit-free inputs with zero mean and unit variance.

6.4 Preprocessing Workflow

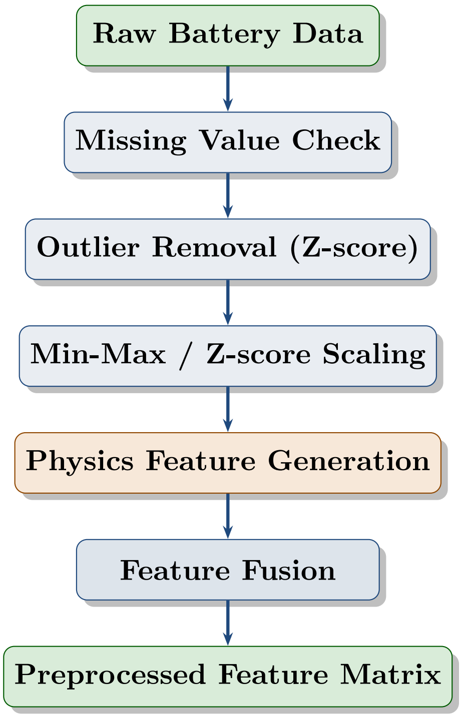

Figure 6.1: Data preprocessing workflow.

Chapter 7

Artificial Neural Network Architecture

7.1 Architecture Overview

The proposed multi-task ANN accepts 14 input features and outputs simultaneous SOH and RUL predictions through a shared latent representation followed by two specialised output heads. Figure 7.1 shows the full architecture.

7.2 Forward Propagation Equations

Let $\mathbf{x} \in \mathbb{R}^{14}$ be the input feature vector. For layer l:

$$\mathbf{z}^{(l)} = \mathbf{W}^{(l)}\mathbf{h}^{(l-1)} + \mathbf{b}^{(l)} \tag{7.1}$$

$$\mathbf{h}^{(l)} = f\big(\mathrm{BN}\big(\mathbf{z}^{(l)}\big)\big) \tag{7.2}$$

where $f(\cdot)$ is the ReLU activation, $\mathrm{BN}(\cdot)$ denotes batch normalisation, and dropout is applied during training. The shared latent representation at layer 3 is:

$$\mathbf{z}_{\text{latent}} = \mathbf{h}^{(3)} \in \mathbb{R}^{128} \tag{7.3}$$

The SOH branch output:

$$\hat{y}_{\text{SOH}} = \mathbf{W}^{(5)}_{\text{SOH}} f\left(\mathbf{W}^{(4)}_{\text{SOH}}\mathbf{z}_{\text{latent}} + \mathbf{b}^{(4)}_{\text{SOH}}\right) + b^{(5)}_{\text{SOH}} \tag{7.4}$$

Analogously for $\hat{y}_{\text{RUL}}$.

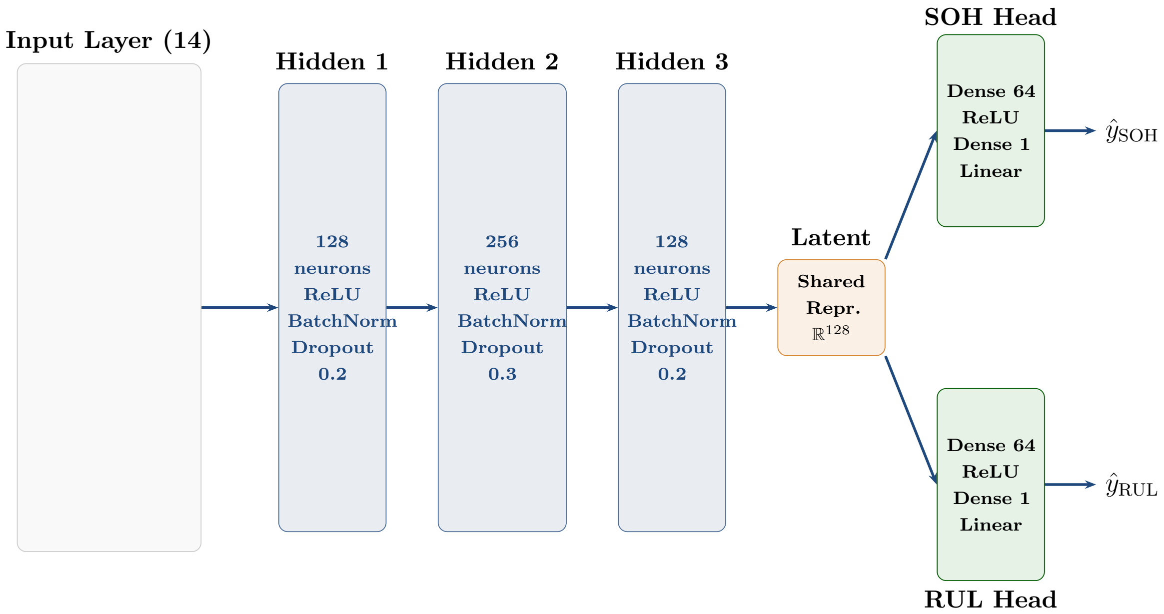

Figure 7.1: Multi-task physics-informed ANN architecture for simultaneous SOH and RUL prediction.

Chapter 8

Activation Functions and Regularisation

8.1 Activation Functions

8.1.1 Rectified Linear Unit (ReLU)

$$f_{\text{ReLU}}(z) = \max(0, z) \tag{8.1}$$

ReLU is selected for hidden layers due to its computational efficiency, absence of vanishing gradient for positive pre-activations, and empirically superior performance in regression tasks.

8.1.2 Leaky ReLU

$$f_{\text{LReLU}}(z) = \begin{cases} z & z > 0 \\ \alpha z & z \leq 0 \end{cases}, \qquad \alpha = 0.01 \tag{8.2}$$

8.1.3 Sigmoid

$$f_{\text{sig}}(z) = \frac{1}{1 + e^{-z}} \tag{8.3}$$

8.1.4 Hyperbolic Tangent

$$f_{\tanh}(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \tag{8.4}$$

8.2 Batch Normalisation

For a mini-batch $\mathcal{B} = \{z_1, \ldots, z_m\}$:

$$\mu_{\mathcal{B}} = \frac{1}{m}\sum_{i=1}^{m} z_i, \qquad \sigma_{\mathcal{B}}^2 = \frac{1}{m}\sum_{i=1}^{m}(z_i - \mu_{\mathcal{B}})^2 \tag{8.5}$$

$$\hat{z}_i = \frac{z_i - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}}, \qquad y_i = \gamma \hat{z}_i + \beta \tag{8.6}$$

where γ (scale) and β (shift) are learnable parameters, and $\epsilon = 10^{-5}$ prevents division by zero. Batch normalisation reduces internal covariate shift, enabling higher learning rates and improving convergence.

8.3 Dropout

During training, each neuron activation h_j is independently set to zero with probability p:

$$\tilde{h}_j = h_j \cdot \text{Bernoulli}(1 - p)/(1 - p) \tag{8.7}$$

During inference, all neurons are active and outputs are used directly without scaling. Dropout rates of $p = 0.2$ (layers 1 and 3) and $p = 0.3$ (layer 2) are used.

8.4 He Weight Initialisation

Weights are initialised using He initialisation, optimal for ReLU activations:

$$\mathbf{W}^{(l)} \sim \mathcal{N}\left(0, \frac{2}{n^{(l-1)}}\right) \tag{8.8}$$

where $n^{(l-1)}$ is the number of input connections to layer l. He initialisation preserves signal variance through deep ReLU networks, preventing vanishing or exploding gradients at initialisation.

Chapter 9

Physics-Informed Loss Function

9.1 Component Loss Functions

9.1.1 SOH Mean Squared Error

$$\mathcal{L}_{\text{SOH}} = \frac{1}{M} \sum_{i=1}^{M} \left(\hat{y}_{\text{SOH}}^{(i)} - y_{\text{SOH}}^{(i)}\right)^2 \tag{9.1}$$

9.1.2 RUL Mean Squared Error

$$\mathcal{L}_{\text{RUL}} = \frac{1}{M} \sum_{i=1}^{M} \left(\hat{y}_{\text{RUL}}^{(i)} - y_{\text{RUL}}^{(i)}\right)^2 \tag{9.2}$$

9.1.3 Physics Consistency Loss

The physics consistency loss penalises deviation from the physics-based estimates:

$$\mathcal{L}_{\text{phys}} = \frac{1}{M} \sum_{i=1}^{M} \left[\left(\hat{y}_{\text{SOH}}^{(i)} - \text{SOH}_{\text{phys}}^{(i)}\right)^2 + \left(\hat{y}_{\text{RUL}}^{(i)} - \text{RUL}_{\text{phys}}^{(i)}\right)^2\right] \tag{9.3}$$

9.1.4 L2 Regularisation

$$\mathcal{L}_{\text{reg}} = \sum_{l} \left\|\mathbf{W}^{(l)}\right\|_F^2 \tag{9.4}$$

9.2 Total Physics-Informed Objective

$$\boxed{\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{SOH}} + \lambda_2 \mathcal{L}_{\text{RUL}} + \lambda_3 \mathcal{L}_{\text{phys}} + \lambda_4 \mathcal{L}_{\text{reg}}} \tag{9.5}$$

9.3 Hyperparameter Selection for Loss Weights

Grid Search. A coarse grid over $\lambda \in \{0.1, 0.5, 1.0, 2.0\}^4$ is evaluated on the validation set.

Bayesian Optimisation. Using Optuna with a Tree-structured Parzen Estimator (TPE) surrogate, the acquisition function is:

$$\alpha_{\text{TPE}}(\boldsymbol{\lambda}) = \frac{p(\boldsymbol{\lambda}|\mathcal{D}_{\text{good}})}{p(\boldsymbol{\lambda}|\mathcal{D}_{\text{bad}})} \tag{9.6}$$

Adaptive Uncertainty Weighting. Following Kendall et al., learnable log-variances $s_i = \log \sigma_i^2$ automatically balance tasks:

$$\mathcal{L}_{\text{unc}} = \sum_{i=1}^{4} \left(\frac{\mathcal{L}_i}{2\sigma_i^2} + \log \sigma_i \right) \tag{9.7}$$

Recommended default values are $\lambda_1 = 1.0$, $\lambda_2 = 0.8$, $\lambda_3 = 0.5$, $\lambda_4 = 10^{-4}$.

Chapter 10

Training Procedure and Hyperparameter Tuning

10.1 Optimiser

The Adam optimiser is employed:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\nabla_\theta \mathcal{L} \tag{10.1}$$

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)(\nabla_\theta \mathcal{L})^2 \tag{10.2}$$

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1-\beta_1^t}, \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1-\beta_2^t} \tag{10.3}$$

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t}+\epsilon}\hat{\mathbf{m}}_t \tag{10.4}$$

with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and initial learning rate $\eta_0 = 10^{-3}$.

10.2 Hyperparameter Configuration

Table 10.1: Training hyperparameter configuration

Hyperparameter	**Default**	**Search range**
Learning rate η_0	1×10^{-3}	$[10^{-4}, 10^{-2}]$
Batch size	32	$\{16, 32, 64, 128\}$
Epochs	300	up to 500
Dropout rate p_1, p_3	0.2	$[0.1, 0.5]$
Dropout rate p_2	0.3	$[0.1, 0.5]$
Weight decay λ_4	10^{-4}	$[10^{-5}, 10^{-3}]$
λ_1 (SOH)	1.0	$[0.5, 2.0]$
λ_2 (RUL)	0.8	$[0.5, 2.0]$
λ_3 (physics)	0.5	$[0.1, 1.0]$
LR decay factor	0.5	–
Early stopping patience	20 epochs	–

10.3 Learning Rate Scheduling

A ReduceLROnPlateau scheduler halves η if the validation loss does not improve over 10 consecutive epochs:

$$\eta_{t+1} = 0.5 \cdot \eta_t \quad \text{if } \mathcal{L}_{\text{val}}^{(t)} \geq \mathcal{L}_{\text{val}}^{(t-k)} \; \forall k \in [1, 10] \tag{10.5}$$

10.4 Training Workflow

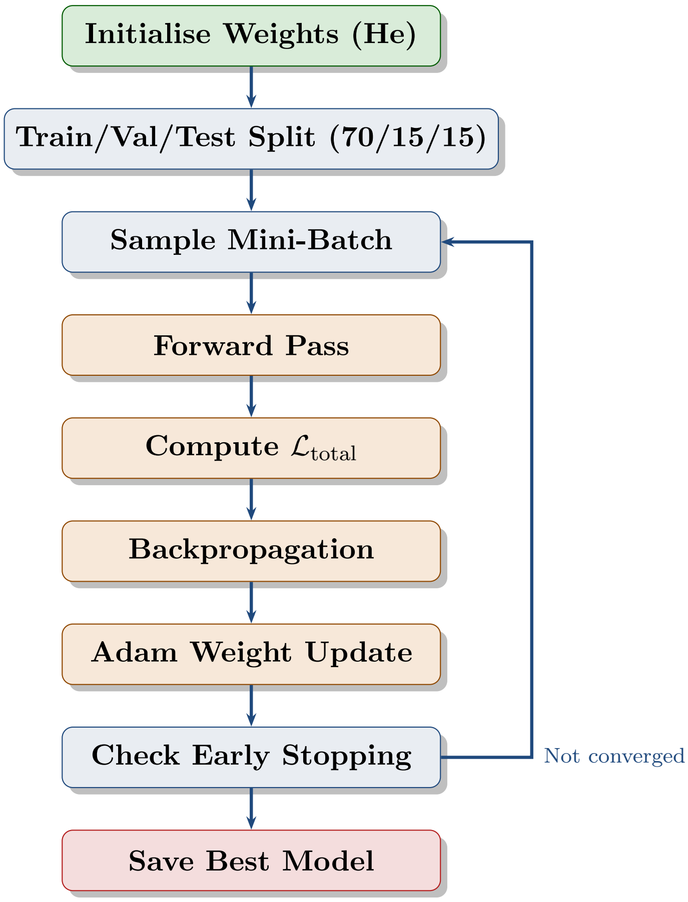

Figure 10.1: Training workflow for the physics-informed ANN.

Chapter 11

Validation Framework and Evaluation Metrics

11.1 Evaluation Metrics

Let $\hat{y}_i$ denote the predicted value and y_i the ground truth for sample i over M samples.

$$\text{MAE} = \frac{1}{M}\sum_{i=1}^{M} |\hat{y}_i - y_i| \tag{11.1}$$

$$\text{MSE} = \frac{1}{M}\sum_{i=1}^{M} (\hat{y}_i - y_i)^2 \tag{11.2}$$

$$\text{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{M} (\hat{y}_i - y_i)^2} \tag{11.3}$$

$$\text{MAPE} = \frac{100\%}{M}\sum_{i=1}^{M} \left|\frac{\hat{y}_i - y_i}{y_i}\right| \tag{11.4}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{M}(\hat{y}_i - y_i)^2}{\sum_{i=1}^{M}(y_i - \bar{y})^2} \tag{11.5}$$

$$R^2_{\text{adj}} = 1 - (1 - R^2)\frac{M-1}{M-p-1} \tag{11.6}$$

where p is the number of input features and $\bar{y}$ is the mean ground truth.

11.1.1 Physics Agreement Score

$$\text{PAS} = 1 - \frac{1}{M}\sum_{i=1}^{M} \left|\hat{y}^{(i)}_{\text{SOH}} - \text{SOH}^{(i)}_{\text{phys}}\right| \tag{11.7}$$

11.1.2 Degradation Trend Consistency Score

$$\text{DTCS} = \frac{1}{M-1}\sum_{i=1}^{M-1} \mathbf{1}\left[\left(\hat{y}^{(i+1)}_{\text{SOH}} - \hat{y}^{(i)}_{\text{SOH}}\right)\cdot\left(y^{(i+1)}_{\text{SOH}} - y^{(i)}_{\text{SOH}}\right) > 0\right] \tag{11.8}$$

11.2 Validation Architecture

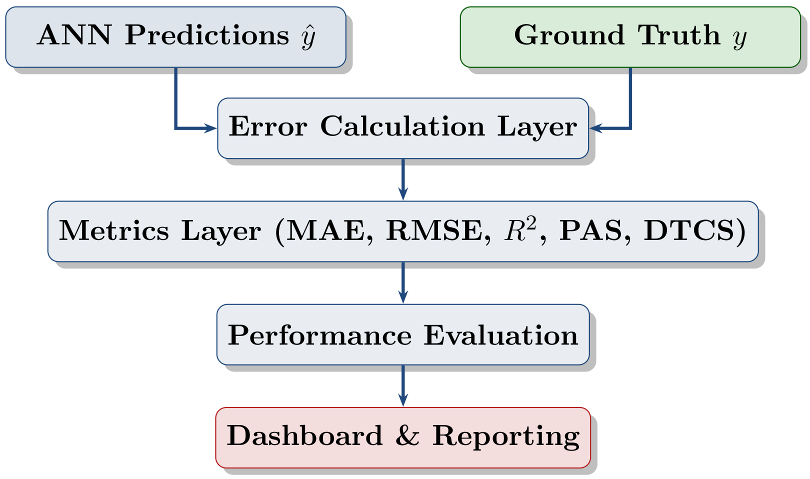

Figure 11.1: Validation framework architecture.

Chapter 12

Hybrid System Architecture

Figure 12.1 presents the complete end-to-end hybrid Physics-Informed ANN system.

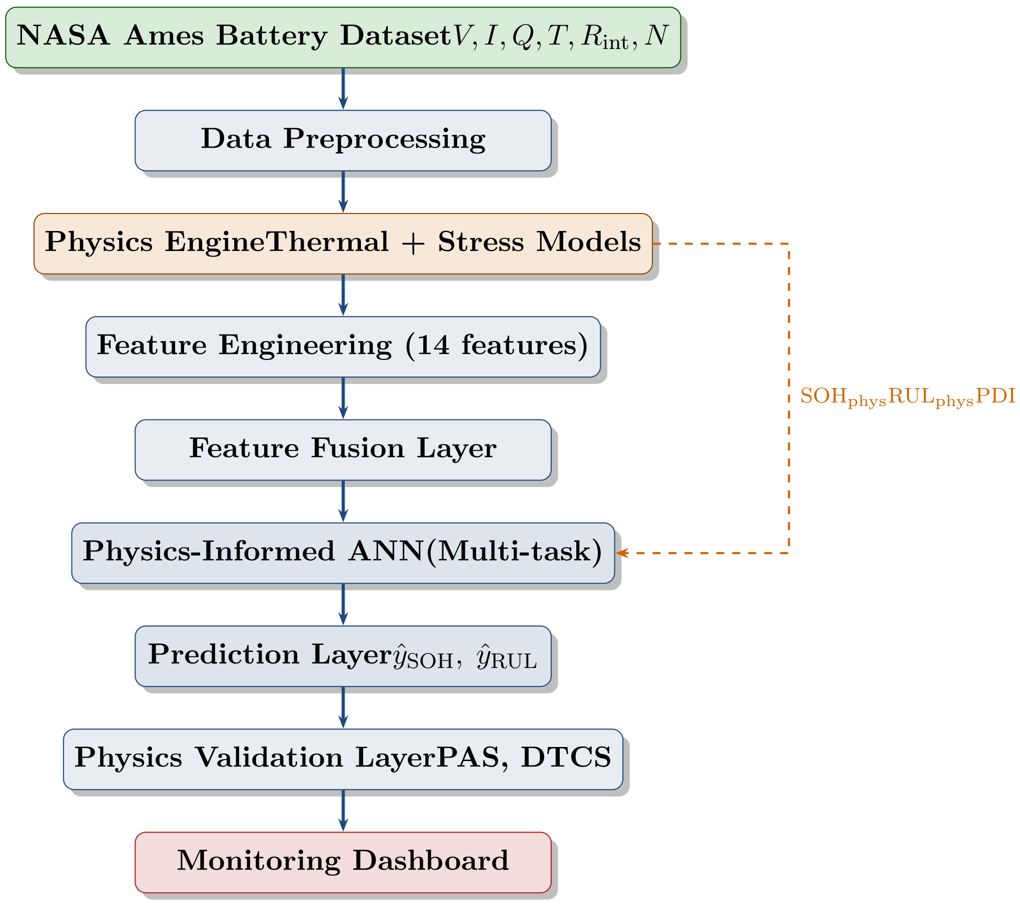

Figure 12.1: Full hybrid Physics-Informed ANN system architecture. The dashed orange arrow represents the physics regularisation signal fed directly into the loss function during training.

Chapter 13

User Workflow and Dashboard

13.1 User Workflow

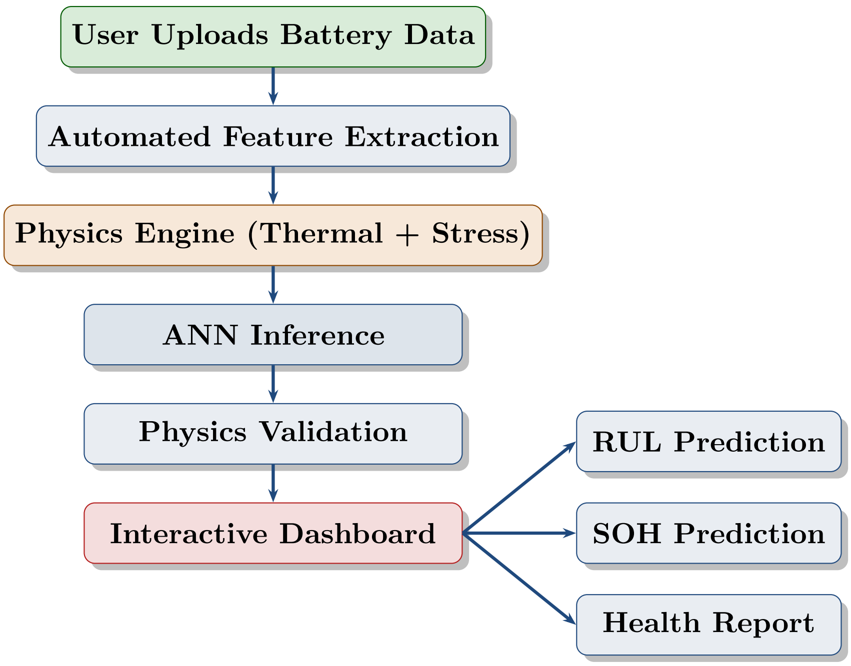

Figure 13.1: End-to-end user workflow.

13.2 Dashboard Components

Table 13.1: Dashboard design — displayed indicators

Panel	**Indicator**	**Action threshold**
Battery Health	Current SOH (%)	Alert if SOH $< 85\%$
Prognosis	Predicted RUL (cycles)	Warning if RUL < 50
Thermal	Thermal gradient ΔT	Alert if $\Delta T > 5\,\mathrm{K}$
Mechanical	Stress Score σ_{norm}	Warning if > 0.7
Model Trust	Physics Agreement Score	Flag if PAS < 0.90
Risk	Battery Risk Level	Low / Medium / High / Critical
Action	Maintenance Recommendation	Text advisory

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